ECE 535 Project

The ECE 535 project is to find a publication or formal project report where pattern recognition and/or machine learning has been deployed to solve a problem and then write a critique on the application of machine learning. A good paper selection should discuss the solution to a specific problem such as detection of heart disease diagnosis from medical data as opposed to a paper which discusses the detection of objects from digital images. General technique papers are not recommended as they will generally not contain enough information to allow for a proper critique of how machine learning was applied to a problem. The paper critique should have the following sections and answer the questions described below in the proposal and final report. The questions must be answered in your own words. Direct copying of important explanations from the source material will be considered plagiarism so be sure to study the source material to sufficient depth so that you obtain a full understanding of what is being performed.

The project is to be performed in groups of two. The source material should either be a conference paper from a top tier journal (IEEE Transactions or equivalent). Acceptable sources for paper are recent journal papers published by the IEEE, ACM, or conference papers that have been published in the last 4 years. Conference papers must be at least 10 pages long from conferences with an acceptance rate of less than 30% to be acceptable.

The final submission is due on the date of the final exam. By February 28th, you will need to submit a one or two page proposal of what paper/report you intend to study for the project.

Proposal breakdown (25% of total project mark), Due February 28th

Max Length: 2 pages with 12 point font. Proposals over this length will only have extra material truncated off before marking.

1. Description of paper to be analyzed (20%)
    a. Provide a copy of the full paper and full proper IEEE citation to where the paper was published.
2. Description of problem that the paper is addressing (40%)
    a. Describe the problem that the paper project is using machine learning/pattern recognition to solve. (10%)
    b. Describe the who will be using the algorithm and how they will be using it. (20%)
        i. Who will use the algorithm? (engineers, medical doctors, the general public?)
        ii. What measurements will the algorithm be using?
        iii. What computing platform must the algorithm run on?
    c. Describe any special considerations of the problem. (10%)
        i. Are their any legal restrictions in the problem domain that the system designers had to consider?
        ii. Is privacy a problem with this problem?
        iii. Are their any ethical considerations with the use of the system?
        iv. Special considerations of the problem such as special characteristics of the sensors, limited availability of data, that make the problem difficult or different from the prior art should be discussed here.
3. Brief overview of the proposed solution and the dataset considerations. (40%)
    a. What algorithms for machine learning is the paper proposing to solve the problem and why are they being selected for this problem? (10%)
    b. What algorithms have previously been proposed/used to solve the described problem or similar problems? (10%)
    c. What improvements does the paper claim in their paper over the previous efforts in this area? This is not just the improved performance metrics but a brief overview of what improvements the authors have added to their algorithm over the prior art. (10%)
    d. What datasets were used to train and evaluate the algorithm? Describe briefly the authors reasoning or justification that these datasets can provide a good measure of real world performance of their proposed method. Comment on if the dataset would give a good representative dataset for the problem to be solved. If the dataset appears deficient you should note this here. (10%)

Final report breakdown (75% of total project mark)

Max Length: 5 pages at 12 point font. Reports over this length will have extra material discarded before marking.

1. Introduction (10%) This should not be a direct copy of the proposal but a shortened version.
    - Describe the problem that the paper/project is using machine learning/pattern recognition to solve.
    - Describe who/what the target audience of the learning algorithm.
        - Who will use the algorithm? (engineers, medical doctors, the general public?)
        - What measurements will the algorithm be using?
        - What computing platform must the algorithm run on?
    - Describe any special considerations of the problem.
        - Are their any legal or other restrictions in the problem domain that the system designers had to consider?
        - Are their any ethical considerations with the use of the system?
2. Data set collection/processing (15%)
    - Describe the source of data sets
    - Comment on how the project ensures that the data set gives a good representation of all classes that are of interest to the system.
        - How did they make sure that the data set gives a full representation of the distribution of sensor measurements/feature values for each class of interest?
        - For data with continuous distributions (e.g. audio), how did they ensure the data represents a reasonable coverage of the whole distribution of the measurements for the classes of interest.
    - Pre-processing
        - Describe the preprocessing performed on the data.
            - Comment on if any outlier removal or normalization were performed on the data.
            - How did the authors determine what preprocessing was to be performed on the data?
            - How did the authors measure the effectiveness of preprocessing?
            - How did the authors ensure that preprocessing did not remove any useful features from the data to be processed?
    - Describe any novel or unique features for the data set collection/processing for this system.
    - Discuss if you feel that the authors used appropriate methods for data set collection/processing. Support your answer.
    - Discuss if you feel that there are any improvements that can be made to the data processing described in the paper/project.
3. Feature selection (15%)
    - How were the relevant features and measurements that were used for the classification/clustering task selected?
    - How was the selection of features validated? How did the authors determine if these features were useful for the task?

- Were any special steps done to ensure that features were invariant to typical transformations of the relevant data? (e.g. features from an image should not be affected by rotation)
- Describe any special or novel features of the feature selection used in this system.
- Discuss if you feel that there are other features that could be used for the task described in the source paper/project. Why do you think that the authors did not use these features?

4. Machine learning/pattern classification algorithm used. (20%)
    - Give a brief overview of the algorithm(s) used in the paper.
    - Describe, in your own words, the justification given by the authors for the algorithm(s) used in the paper.
        - Discuss if you agree with the authors selection of algorithm. Support your answer based on concepts covered in the class.
    - Describe how the authors selected the parameters of the learning algorithm used.
        - This could be number of layers/nodes of the neural network, activation functions in the nodes, amount of feedback used, types of layers deployed in the network, etc.
        - Discuss if you agree with their selection of parameters. Do you think that their system performance good be improved.
    - Discuss the selection of parameters and algorithm in terms of the computational platform that the system is likely to be implemented on.
        - For example, if the system is to be put in an embedded device is the algorithm likely to fit on such a device.
    - Describe any special or novel features of the machine learning/classification algorithm used in this system.
    - Discuss if you agree with the selection of algorithms used in the paper. Give reasons for your answer.
5. Performance Evaluation (20%)
    - Discuss how the authors tested the performance of their system.
    - What evaluation set(s) were used to test the algorithm.
    - How did the authors determine that their evaluation sets were independent of the training sets?
    - Did the authors require any other sets other than training and evaluation sets? For example, did they need a 'test set' to perform evaluations to help select which of several models to use in the learning algorithm? Did they use cross-validation to learn model parameters or for other purposes? If cross-validation was used, describe the cross validation and discuss if the cross-validation methods seems appropriate.
    - How did the authors determine if their evaluation would actually give a good measure of the general performance of their algorithm?
    - Do the authors describe what future evaluations of their algorithm would be needed before general deployment can be done?
    - Describe any special or novel features of the performance evaluation used in this system.

- Discuss if you feel that the authors properly evaluated the performance of their systems or there are other methods that could give more accurate measures of the system's performance on general data.

6. Conclusions (10%)
    - Give your final conclusions about the system design or presentation in the paper.
    - Do you agree with the conclusions drawn by the authors of the paper/project.
    - If you were to continue the project or to do the project yourself, what changes would you make?
    - Do you think that the system designed is suitable for the stated purpose?
    - Are their any open questions that you think need to be answered before the system can be widely deployed?
        - If you answer yes to this question, describe briefly what these questions are.

Note that there is an additional 10% for presentation. The language use in your report should be good with acceptable use of grammar and spelling. The text should be pleasant to view with appropriate use of spacing and headers. Figures should be easy to read with appropriate captions and axis labels. Text taken from other sources should be appropriately cited.

Note: The report is to be your evaluation of the paper's use of pattern recognition methods; it is not to be a restatement of the source papers' methodology and conclusions. Direct copying of material from the source paper or asking AI tools such as ChatGPT to restate the source paper content are violations the University of Victoria academic integrity policy and will be dealt with accordingly.

Marking Breakdown

The breakdown of the marks for the project is given below:

Marking Section	Weight
Presentation	10%
Introduction	10%
Data set collection/processing	15%
Feature selection	15%
Machine learning/pattern classification algorithm	20%
Performance Evaluation	20%
Conclusions	10%

For the presentation, the marking guide is:

Points	Comments
0-2	Extremely poor presentation. A large number of grammar and spelling mistakes. Missing important headings. Relevant information is hard to locate; present in the wrong sections or in sections whose heading has no correspondence to that information. Tables and graphs do not help with understanding of material. Presentation issues make most important points of project difficult to find or understand.
3-4	Poor presentation. Headings are used to organize most material in a useful manner. Presentation issues make many important points, such as the description of training set selection method and algorithm description, difficult to find or understand.
5-6	Slightly flawed presentation. Heading used to organize information to organize information. Table and graphs used to presentations of material in a useful way. Either one main point is presented badly or several minor points are difficult to find or understand due to presentation issues.
7-8	Good presentation. Headings are used to organize important information. Table and graphs used to help presentation of material in a useful way. Presentation makes minor but not irrelevant point difficult to understand.
9-10	Excellent presentation. A small number or no grammar or spelling mistakes. Headings make information easy to find. Tables and graphs aid with understanding of material. Excellent use of paragraphs to organize material. Each section is organized in a manner that makes explanations and presentation easy to follow.

Each the other sections are marked as follows (scale to from 0-10 to grade in section above)

Points	Comments
0-2	No information on area or only a superficial coverage of area. Most points are not addressed by report.
3-4	Poor discussion of area. At least two important points are not discussed.
5-6	Slightly flawed discussion. Most important points are discussed but important information needed to understand, or support points is missing.

7-8	Good discussion. All important points are discussed with only a few minor points possibly being absent which does not affect main discussion. Main points are discussed but details are missing.
9-10	Excellent discussion. All important points are clearly discussed with possibly only one minor point missing.